1. Replicator dynamics

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x}))$$

2. Mutual invasion conditions

At $\mathbf{e}_i$, can j invade?

$$\pi_j(\mathbf{e}_i) > \pi_i(\mathbf{e}_i)$$

At $\mathbf{e}_j$, can i invade?

$$\pi_i(\mathbf{e}_j) > \pi_j(\mathbf{e}_j)$$

3. Multiplicative payoffs

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$$

Preserves invasion conditions and allows ratios

4. SCT decomposition

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$$

5. Coexistence criterion

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}$$

Stabilizing niche differences must overcome average fitness differences

6. Niche-fitness difference space